

These are common networking commands used for diagnostics and troubleshooting. Here's a brief explanation of each:

1. **netstat (Network Statistics)**

- Displays network connections (both incoming and outgoing), routing tables, and network interface statistics.
- Example:

```
netstat -an
```

(Shows all active connections and listening ports)

Proto	Local Address	Foreign Address	State
TCP	0.0.0.0:135	0.0.0.0:0	LISTENING
TCP	0.0.0.0:445	0.0.0.0:0	LISTENING
TCP	0.0.0.0:3306	0.0.0.0:0	LISTENING
TCP	0.0.0.0:5040	0.0.0.0:0	LISTENING
TCP	0.0.0.0:7070	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49664	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49665	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49666	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49667	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49668	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49670	0.0.0.0:0	LISTENING
TCP	127.0.0.1:54575	127.0.0.1:4843	SYN_SENT
TCP	172.16.6.91:139	0.0.0.0:0	LISTENING
TCP	172.16.6.91:52632	20.198.118.190:443	ESTABLISHED
TCP	172.16.6.91:52633	20.198.118.190:443	ESTABLISHED
TCP	172.16.6.91:52753	20.198.118.190:443	ESTABLISHED
TCP	172.16.6.91:52825	142.251.10.188:5228	ESTABLISHED
TCP	172.16.6.91:53347	52.108.44.3:443	ESTABLISHED
TCP	172.16.6.91:53395	4.224.93.21:443	ESTABLISHED
TCP	172.16.6.91:53417	52.98.58.50:443	ESTABLISHED
TCP	172.16.6.91:53418	203.23.178.57:443	ESTABLISHED
TCP	172.16.6.91:53437	4.186.14.122:443	ESTABLISHED
TCP	172.16.6.91:54076	8.8.8.8:443	ESTABLISHED
TCP	172.16.6.91:54080	52.98.88.242:443	ESTABLISHED
TCP	172.16.6.91:54092	52.108.8.12:443	ESTABLISHED
TCP	172.16.6.91:54096	52.108.36.33:443	ESTABLISHED
TCP	172.16.6.91:54255	172.64.151.4:443	TIME_WAIT
TCP	172.16.6.91:54299	104.18.33.45:443	ESTABLISHED
TCP	172.16.6.91:54302	104.16.79.73:443	ESTABLISHED
TCP	172.16.6.91:54314	172.64.155.209:443	ESTABLISHED
TCP	172.16.6.91:54315	34.128.128.0:443	ESTABLISHED
TCP	172.16.6.91:54316	151.101.64.176:443	ESTABLISHED
TCP	172.16.6.91:54323	34.128.128.0:443	ESTABLISHED
TCP	172.16.6.91:54324	172.64.155.209:443	ESTABLISHED
TCP	172.16.6.91:54332	172.64.146.98:443	ESTABLISHED
TCP	172.16.6.91:54340	172.64.155.209:443	ESTABLISHED
TCP	172.16.6.91:54343	172.64.151.4:443	ESTABLISHED
TCP	172.16.6.91:54351	35.244.251.182:443	ESTABLISHED

```

TCP    [::]:445          [::]:0            LISTENING
TCP    [::]:3306         [::]:0            LISTENING
TCP    [::]:7070         [::]:0            LISTENING
TCP    [::]:49664        [::]:0            LISTENING
TCP    [::]:49665        [::]:0            LISTENING
TCP    [::]:49666        [::]:0            LISTENING
TCP    [::]:49667        [::]:0            LISTENING
TCP    [::]:49668        [::]:0            LISTENING
TCP    [::]:49670        [::]:0            LISTENING
TCP    [::1]:49669       [::]:0            LISTENING
UDP    0.0.0.0:5050      *:.*
UDP    0.0.0.0:5353      *:.*
UDP    0.0.0.0:5353      *:.*
UDP    0.0.0.0:5353      *:.*
UDP    0.0.0.0:5355      *:.*
UDP    0.0.0.0:50001     *:.*
UDP    0.0.0.0:50082     *:.*
UDP    127.0.0.1:1900    *:.*
UDP    127.0.0.1:49664   *:.*
UDP    127.0.0.1:64376   *:.*
UDP    172.16.6.91:137   *:.*
UDP    172.16.6.91:138   *:.*
UDP    172.16.6.91:1900  *:.*
UDP    172.16.6.91:64375 *:.*
UDP    [::]:5353         *:.*
UDP    [::]:5353         *:.*
UDP    [::]:5355         *:.*
UDP    [::]:50082        *:.*
UDP    [::1]:1900        *:.*
UDP    [::1]:64374       *:.*
UDP    [fe80::74a8:b343:87d2:9c3f%13]:1900 *:.*
UDP    [fe80::74a8:b343:87d2:9c3f%13]:64373 *:.*

```

Local Address vs. Foreign Address in `netstat` Output

In the `netstat -an` output, you see **Local Address** and **Foreign Address** columns. Here's what they mean:

1. Local Address

- Represents **your computer's IP address and port number**.
- This is the network interface that is either **listening for incoming connections** or **sending data to a remote system**.

◆ Example:

```

TCP    172.16.6.91:52632    20.198.118.190:443    ESTABLISHED

```

- **172.16.6.91:52632** (Local Address)
 - 172.16.6.91 → Your computer's private IP address.
 - 52632 → The dynamically assigned local port used for this connection.

2. Foreign Address

- Represents **the remote computer's IP address and port number** that your system is communicating with.
- If the connection is **outgoing**, the foreign address is the destination.
- If the connection is **incoming**, the foreign address is the source.

♦ Example:

```
TCP      172.16.6.91:52632      20.198.118.190:443      ESTABLISHED
```

- **20.198.118.190:443** (Foreign Address)
 - 20.198.118.190 → The remote server's IP address.
 - 443 → The port number (HTTPS, meaning this is a secure web connection).

Your `netstat -an` output provides a snapshot of active network connections and listening ports on your system. Here's a breakdown of what it means:

Key Sections in Your Output:

1. Listening Ports (**LISTENING** state)

- Ports such as 135, 445, 3306, 5040, 7070, etc., are open and waiting for incoming connections.
- **Common Ports Found:**
 - 135 (Microsoft RPC)
 - 445 (SMB over TCP)
 - 3306 (MySQL database service)
 - 7070 (Commonly used for RTSP streaming)

2. Active Connections (**ESTABLISHED** state)

- These are ongoing communication sessions between your computer and remote servers.
- Most connections are over port **443 (HTTPS)**, indicating secure web traffic to external IPs.
- **Examples:**
 - 172.16.6.91:52632 -> 20.198.118.190:443 (Connected to a remote HTTPS server)
 - 172.16.6.91:54076 -> 8.8.8.8:443 (Connection to Google DNS over HTTPS)

3. Connections in **TIME_WAIT** state

- These are connections that recently closed but are still waiting to be fully terminated.
- You have many `TIME_WAIT` connections, which is normal if your system is actively browsing or using cloud services.

4. **UDP Connections**

- UDP ports don't show a "state" since UDP is connectionless.
 - Ports like 5353, 1900, and 50082 indicate services like:
 - 5353 (mDNS/Bonjour for service discovery)
 - 1900 (UPnP device discovery)
-

2. **ping (Packet Internet Groper)**

- Tests connectivity between two network devices using ICMP Echo Requests.
- Example:

```
nginx
ping google.com
```

(Sends packets to Google's server and checks response time)

3. **telnet (Teletype Network Protocol)**

- Used to connect to remote devices over a network using TCP. Commonly used to test open ports.
- Example:

```
telnet 192.168.1.1 80
```

(Attempts to connect to IP 192.168.1.1 on port 80)

Telnet (Teletype Network Protocol)

Telnet is a network protocol used to **connect to remote devices** over a network using **TCP (Transmission Control Protocol)**. It allows users to **execute commands** on a remote machine as if they were physically present at that machine.

However, Telnet is **insecure** because it transmits data, including passwords, in **plain text** (unencrypted). Because of this, it has mostly been replaced by **SSH (Secure Shell)** for remote management.

How Telnet Works

- Establishes a **TCP connection** to a remote system.

- Sends commands in **clear text** (not encrypted).
 - Used for **network troubleshooting**, testing open ports, and connecting to old network devices (e.g., routers, switches, legacy servers).
-

Example: Testing an Open Port with Telnet

```
telnet 192.168.1.1 80
```

◆ Explanation:

- 192.168.1.1 → IP address of the remote system.
- 80 → Port number (commonly used for HTTP web traffic).

✦ If the connection is successful:

- The screen will go blank or show a banner, indicating that the port is open and responding.
- This means the remote machine is accepting connections on port 80.

✦ If the connection fails:

- You will see an error like `Could not open connection to the host, on port 80: Connect failed`, which means the port is closed or blocked.
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C:\Users\Dell>telnet mail.example.com 25 'telnet' is not recognized as an internal or external command, operable program or batch file.

It looks like Telnet is **not installed** on your Windows system. By default, Telnet is **disabled** in modern versions of Windows (Windows 10, 11, and Windows Server). You need to manually enable it.

🔧 How to Enable Telnet in Windows

Method 1: Enable via Command Prompt (Admin)

1. Open **Command Prompt as Administrator**:
 - Press Win + R, type `cmd`, and press Ctrl + Shift + Enter.
2. Run the following command to install Telnet:

```
dism /online /Enable-Feature /FeatureName:TelnetClient
```

3. Wait for the installation to complete.

4. Try running Telnet again:

```
telnet mail.example.com 25
```

Method 2: Enable via Control Panel

1. Press **Win + R**, type **optionalfeatures**, and press **Enter**.
2. Scroll down and **check** the box for **Telnet Client**.
3. Click **OK**, then wait for the installation to complete.
4. Open **Command Prompt** and type:

```
telnet
```

If Telnet is installed, you will enter the Telnet command mode.

Other Uses of Telnet

1. **Check if an SMTP (Mail) server is running:**

```
telnet mail.example.com 25
```

- If successful, you'll see a response from the mail server.

2. **Access a Remote Device (like a switch or router):**

```
telnet 192.168.1.100
```

- If a Telnet service is running, it will ask for a username and password.
-

Why Telnet is Insecure & Alternatives

Security Risk:

- Since Telnet **does not encrypt** data, anyone on the network can capture sensitive information (e.g., passwords).
- **Better Alternative:** Use **SSH (Secure Shell)** instead:

```
ssh user@192.168.1.1
```

4. `ftp` (File Transfer Protocol)

- Transfers files between a local and remote system.
- Example:

```
ftp example.com
```

(Starts an FTP session with `example.com`)

FTP (File Transfer Protocol)

FTP (File Transfer Protocol) is a standard network protocol used to transfer files between a **local** computer and a **remote** server over a TCP-based network, like the Internet.

How FTP Works

- **Client-Server Model:**
 - An FTP client (your computer) connects to an FTP server.
 - The server either allows anonymous access or requires authentication (username & password).
 - Files can be **uploaded, downloaded, deleted, or listed** from the remote system.
 - **Control & Data Channels:**
 - **Control Channel (Port 21):** Manages commands and responses.
 - **Data Channel (Port 20 or dynamic):** Transfers actual files.
-

FTP Modes

1. **Active Mode**
 - The **server** actively establishes the data connection.
 - Less firewall-friendly due to server-initiated connections.
 2. **Passive Mode (PASV)**
 - The **client** initiates the data connection.
 - Works better with firewalls and NAT (used in most modern setups).
-

Example of FTP Usage

1. Connect to an FTP Server

```
ftp example.com
```

- You will be prompted for a **username** and **password**.

2. List Files on the Server

```
ls
```

- Displays files and directories on the remote server.

3. Download a File

```
get filename.txt
```

- Downloads `filename.txt` from the server to your local machine.

4. Upload a File

```
put localfile.txt
```

- Uploads `localfile.txt` from your local system to the FTP server.

5. Exit the FTP Session

```
bye
```

- Closes the FTP connection.

5. traceroute (or tracert on Windows)

- Traces the route packets take to a destination and displays each hop along the way.
- Example:

```
tracert google.com
```

(Shows the path packets take to reach Google's servers)

Hops Breakdown

Hop	Time 1	Time 2	Time 3	Router/IP
1	4 ms	2 ms	1 ms	172.16.1.1
2	*	25 ms	29 ms	undefined.hostname.localhost [202.160.171.169]
3	36 ms	26 ms	28 ms	72.14.194.150
4	27 ms	*	*	172.253.68.93
5	*	27 ms	*	142.251.54.101
6	20 ms	20 ms	18 ms	del11s18-in-f14.1e100.net [142.250.193.238]

Step-by-Step Explanation

- Hop 1 (172.16.1.1)**
 - This is likely your **router/gateway** (private IP).
 - Low response time (1-4ms) suggests it's close.
- Hop 2 (202.160.171.169)**
 - This is probably your **ISP's router**.
 - The `undefined.hostname.localhost` means the ISP hasn't assigned a name to this IP.
 - One request timed out (*), which means no response for that attempt.
- Hop 3 (72.14.194.150)**
 - This belongs to **Google's network**.
 - The response times are consistent (~26-36 ms), indicating a good connection.
- Hop 4 (172.253.68.93) & Hop 5 (142.251.54.101)**
 - These are **Google's backbone servers**.
 - Some packets were dropped (*), but the route is still working.
- Hop 6 (142.250.193.238)**
 - This is the **Google server** in Delhi (as indicated by `del11s18`).
 - The response time (~18-20ms) is good.

Key Observations

✓ Packet Loss (*)

- Some requests timed out at hops 2, 4, and 5.
- This can happen if routers **prioritize actual traffic over ICMP requests** (which `tracert` uses).
- Not a major issue unless it happens consistently.

✓ Latency (ms values)

- Latency increases slightly with each hop but remains low (~20-36ms).
- A high jump in latency at any hop could indicate a **network bottleneck**.

✓ Final Destination Reached (Trace complete.)

- Your computer successfully reached **Google's server** at 142.250.193.238.

What Can You Do With This Information?

- If some hops have **very high latency or 100% packet loss**, it may indicate **network congestion** or **ISP issues**.
- If the **final hop has a high response time**, it may mean **Google's server is slow or far away**.
- If you get `Request timed out` for all hops, your **network is likely blocking ICMP packets**.

Why Are There Three Different Times?

- `tracert` **sends three probes (ICMP packets)** per hop to get an average response time.
- The times may **differ** due to:
 1. **Network congestion** (some packets may take slightly different paths)
 2. **Router load balancing** (packets may take alternate routes)
 3. **Processing delay** (routers may be handling other traffic)
 4. **Packet prioritization** (ICMP packets may be deprioritized)

How to Interpret Time Differences

Time Differences	Meaning
All three values are similar	Stable connection, minimal variation
One value is significantly higher	Temporary delay or congestion
All values are high and increasing at each hop	Possible network bottleneck or long-distance routing
Values fluctuate a lot across hops	Possible unstable network

If you see a **huge jump in latency at a certain hop** and it continues beyond that, it could indicate a **network slowdown** or an issue with a specific router.